

► MICROBIOLOGY—MYCOLOGY

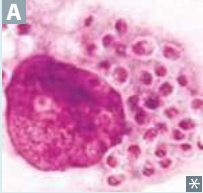
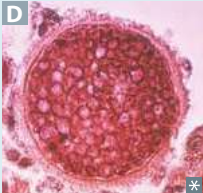
Systemic mycoses

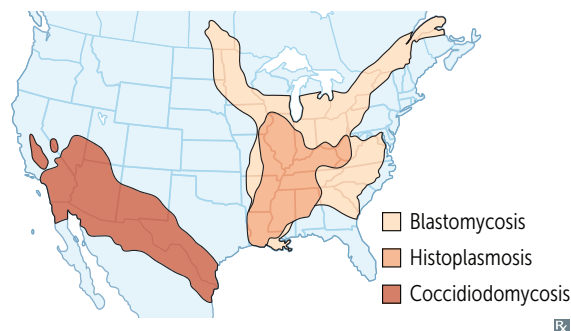
All of the following can cause pneumonia and can disseminate.

All are caused by dimorphic fungi: **cold** (20°C) = **mold**; **heat** (37°C) = **yeast**. Only exception is *Coccidioides*, which is a spherule (not yeast) in tissue.

Systemic mycoses can form granulomas (like TB); cannot be transmitted person-to-person (unlike TB).

Treatment: fluconazole or itraconazole for **local** infection; amphotericin B for **systemic** infection.

DISEASE	ENDEMIC LOCATION	PATHOLOGIC FEATURES	UNIQUE SIGNS/SYMPTOMS	NOTES
Histoplasmosis 	Mississippi and Ohio River Valleys	Macrophage filled with <i>Histoplasma</i> (smaller than RBC) A Tuberculate macroconidia on culture	Palatal/tongue ulcers, splenomegaly, pancytopenia, erythema nodosum	Histo hides (within macrophages) Associated with bird or bat droppings (eg, caves) Diagnosis via urine/serum antigen
Blastomycosis 	Eastern and Central US, Great Lakes	Broad -based budding of <i>Blastomyces</i> (same size as RBC) B	Inflammatory lung disease Disseminates to bone/skin (verrucous lesions C , may mimic SCC).	Blasto buds broadly 
Coccidioidomycosis 	Southwestern US, California	Spherule filled with endospores of <i>Coccidioides</i> (much larger than RBC) D	Disseminates to bone/skin Erythema nodosum (desert bumps) or multiforme Arthralgias (desert rheumatism) Can cause meningitis	Associated with dust exposure in endemic areas (eg, archeological excavations, earthquakes)
Para-coccidioidomycosis 	Latin America	Budding yeast of <i>Paracoccidioides</i> with “ captain’s wheel ” formation (much larger than RBC) E	Similar to blastomycosis, males > females	Paracoccidio parasails with the captain’s wheel all the way to Latin America



Opportunistic fungal infections

Candida albicans

alba = white. Dimorphic; forms pseudohyphae and budding yeasts at 20°C **A**, germ tubes at 37°C **B**.

Systemic or superficial fungal infection. Causes oral **C** and esophageal thrush in immunocompromised (neonates, steroids, diabetes, AIDS), vulvovaginitis (diabetes, use of antibiotics), diaper rash, infective endocarditis (people who inject drugs), disseminated candidiasis (especially in neutropenic patients as host defense relies on phagocytes), chronic mucocutaneous candidiasis.

Treatment: oral fluconazole/topical azoles for vaginal; nystatin, azoles, or, rarely, echinocandins for oral; fluconazole, echinocandins, or amphotericin B for esophageal or systemic disease.

Aspergillus fumigatus

Acute angle (45°) **D** branching of **se**ptate hyphae.

Causes invasive aspergillosis in immunocompromised patients, especially those with neutrophil dysfunction (eg, chronic granulomatous disease) because *Aspergillus* is catalase ⊕.

Can cause aspergillomas **E** in pre-existing lung cavities, **especially after TB infection**.

Some species of *Aspergillus* produce aflatoxins (induce TP53 mutations leading to hepatocellular carcinoma).

Treatment: voriconazole or echinocandins (2nd-line).

Allergic bronchopulmonary aspergillosis (ABPA)—hypersensitivity response to *Aspergillus* growing in lung mucus. Associated with asthma and cystic fibrosis; may cause bronchiectasis and eosinophilia.

Cryptococcus neoformans

5–10 μm with narrow budding. Heavily encapsulated yeast. Not dimorphic. ⊕ PAS staining.

Found in soil, pigeon droppings. Acquired through inhalation with hematogenous dissemination to meninges. Highlighted with India ink (clear halo **F**) and mucicarmine (red inner capsule **G**).

Latex agglutination test detects polysaccharide capsular antigen and is more sensitive and specific. Causes cryptococcosis, which can manifest with meningitis, pneumonia, and/or encephalitis (“soap bubble” lesions in brain), primarily in immunocompromised.

Treatment: amphotericin B + flucytosine followed by fluconazole for cryptococcal meningitis.

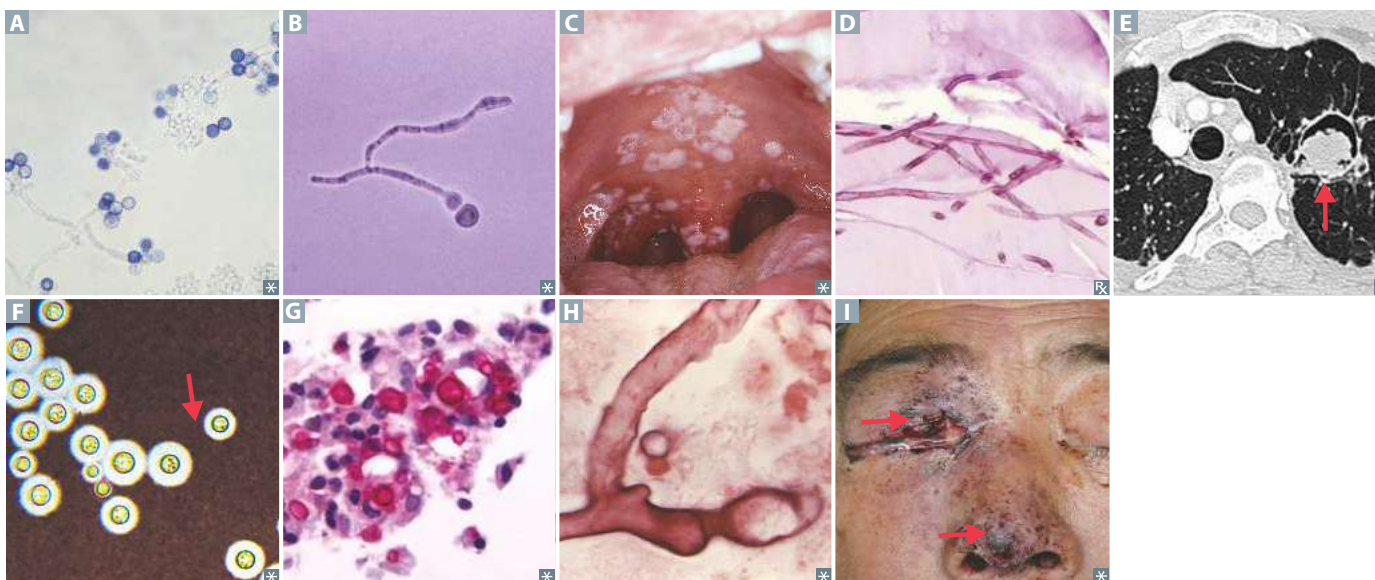
Mucor and Rhizopus spp

Irregular, broad, nonseptate hyphae branching at wide angles **H**.

Causes mucormycosis, mostly in patients with DKA and/or neutropenia (eg, leukemia). Inhalation of spores → fungi proliferate in blood vessel walls, penetrate cribriform plate, and enter brain.

Rhinocerebral, frontal lobe abscess; cavernous sinus thrombosis. Headache, facial pain, black necrotic eschar on face **I**; may have cranial nerve involvement.

Treatment: surgical debridement, amphotericin B or isavuconazole.



Pneumocystis jirovecii

Yeastlike fungus that causes *Pneumocystis* pneumonia (PCP), a diffuse interstitial pneumonia **A**. Most infections are asymptomatic. Immunosuppression (eg, AIDS) predisposes to disease. Diffuse, bilateral ground-glass opacities on chest imaging, with pneumatoceles **B**. Diagnosed by bronchoalveolar lavage or lung biopsy. Disc-shaped yeast seen on methenamine silver stain of lung tissue **C** or with fluorescent antibody.

Treatment/prophylaxis: TMP-SMX, pentamidine, dapsone (prophylaxis as single agent, or treatment in combination with TMP), atovaquone. Start prophylaxis when CD4+ cell count drops to < 200 cells/mm³ in people living with HIV.

***Sporothrix schenckii***

Causes sporotrichosis. Dimorphic fungus. Exists as a **cigar**-shaped yeast at 37 °C in the human body and as hyphae with spores in soil (conidia). Lives on vegetation. When spores are traumatically introduced into the skin, typically by a thorn ("**rose gardener's** disease"), causes local pustule or ulcer with nodules along draining lymphatics (ascending lymphangitis **A**). Disseminated disease possible in immunocompromised host.

Treatment: itraconazole or **pot**assium iodide (only for cutaneous/lymphocutaneous).

Think of a **rose gardener** who smokes a **cigar** and **pot**.